

# SHAURYA BHATNAGAR

B.Tech Mechanical Engineering | 2nd Year, 4th Semester

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## TECHNICAL SKILLS

|                           |                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>CAD / Design</b>       | <b>ANSYS(Fluent, CFX, Mechanical, workbench)</b> SolidWorks (Advanced), Fusion 360, AutoCAD                 |
| <b>Simulation</b>         | ANSYS Fluent/Mechanical/ROCKY, SolidWorks FEA                                                               |
| <b>Manufacturing</b>      | Additive Manufacturing, FDM 3D Printing, UV-Hardening Epoxy Composites                                      |
| <b>Domains</b>            | Certified for Supersonics, Aerodynamics, UAV Systems, Rocket Design, Parametric Design                      |
| <b>Tools</b>              | <b>ANSYS, ABAQUS</b> , SolidWorks Equations-Driven Modelling, Parametric Assembly Design                    |
| <b>Currently Learning</b> | Simulia Abaqus (FEA/Structural Simulation), Python for ANSYS/Abaqus Scripting, Solidworks Etiquette (ASME). |

## OBJECTIVE

Keen to learn, Motivated, Curious 2nd-year Mechanical Engineering student with applied experience in aerodynamic design, CAD modelling, flow simulation, and UAV development. Seeking an internship in defence technology or aerodynamic research at DRDO/NAL/ISRO/HAL or IIT research labs to contribute to cutting-edge aerospace and defence systems.

## AREAS OF INTEREST

**HYPERSONICS, SUPERSONICS** | Aerodynamics & CFD UAV | UAS Systems | Rocket Nozzle | Deployable Space Structures | Parametric & Generative Design | **Defence Technology**

## KEY PROJECTS (TOP 10 PROJECTS)

### Supersonic/Hypersonic Flow on Wedge **FOR MISSILE TECH** *[curiosity/interest]*

- ▶ 30 Deg Wedge Simulation certified by Ansys Innovation Courses.
- ▶ Designed a wedge to simulate airflow on **BRAHMOS**, TRBM 155HG, Kh-31.
- ▶ Understanding of RAMJET compression.

### Static Pressure analysis for Submarine Vehicles *[curiosity/interest]*

- ▶ Dual depth analysis for distinct altitudes.
- ▶ Prediction of Variable Surfaces on Variable Depths.
- ▶ Planning to "Design" submarine drones.

### DA-LAVAL Nozzle Study *[Self-Initiated]*

- ▶ Used CutPlots and Surface Plot to understand the Flow of gases with **High Mach Number** in a perfect nozzle.

### Subsonic E-Class Model Rocket *[Team Project]*

- ▶ Designed, built,(yet to be launched) subsonic E-class rocket: covering propulsion, structural integrity, and flight dynamics.
- ▶ Applied aerodynamic stability principles ( $C_p$  vs  $C_g$  analysis) during design phase.
- ▶ Team engineering a successfully deployable full parachute recovery system, ensuring controlled descent and vehicle recovery post-apogee

### Fixed-Wing High-Wing RC UAV + Aerodynamic Flow Analysis *[Community-Initiated -> Personal Project]*

- ▶ Originally began as a community project; independently continued and expanded into a personal aerodynamic research build.
- ▶ Fabricated an ultra-low-cost fixed-wing high-wing RC UAV from scratch.
- ▶ Conducted 15+ flow simulations in SolidWorks Flow Simulation to identify vortex sources, turbulence regions, and aerodynamic instabilities, Cyclically redesigned wing and fuselage geometry to improve lift-to-drag ratio and flight stability..

### Laser Communication System *[Self-Initiated]*

- ▶ Designed and built a functional laser-based communication system demonstrating line-of-sight optical data transmission.
- ▶ Allows for Spatial Isolation prevents RF eavesdropping using LOS only characteristics. *Inherently secure* while minimising overhead

### Miura-Ori Motion Study for Space-Deployable Solar Panels *[Independent Research/Study]*

- ▶ Studied the kinematics of Miura-Ori origami fold patterns for application in deployable space structures.
- ▶ Analysed geometric efficiency gains for solar panel packaging and deployment in space missions.

#### **Parametric Gearbox: SolidWorks Equations-Driven** [Self-Initiated]

- ▶ Developed an **industry-ready parametric gearbox** assembly in SolidWorks using the **Equations tab**, fully driven by a GearboxConfiguration.txt file.
- ▶ User inputs module and number of teeth for each of the 4 gears; all geometry auto-scales! **ZERO** manual redrawing required.

#### **AutoGEAR: Standalone Parametric Gear Generator** [Self-Initiated]

- ▶ Extracted the gear design logic into a standalone parametric gear generator tool for rapid single gear modelling.

#### **Hybrid Additive Manufacturing 3D Print + UV Epoxy Composites** [Self-Initiated/For Team Project]/(UnDocumented)

- ▶ Developed a manufacturing technique combining FDM 3D printing with UV-hardening epoxy to create high-strength structural components.
- ▶ Applied this technique to fabricate a drop-proof nosecone for the E-Class rocket, a direct aerospace application of composite additive manufacturing. Explored composite reinforcement strategies for lightweight, load-bearing aerospace structures.

## **EDUCATION**

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### **B.Tech – Mechanical Engineering** [2024 – 2028 (Expected)]

CSJM University (Chhatrapati Shahu Ji Maharaj University), Kanpur | CGPA: 7.3 / 10.0 | 2nd Year, 4th Semester

### **Class XII – Zennext Public School, Ghaziabad** [2024]

CBSE | 86%

### **Class X – Zennext Public School, Ghaziabad** [2022]

CBSE | 86%

## **CURRENTLY LEARNING**

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- ▶ ANSYS Fluent: Advanced CFD solver for compressible/incompressible flow, turbulence modelling, and aerodynamic analysis
- ▶ Simulia Abaqus: Finite Element Analysis (FEA) for structural, thermal, and dynamic simulation of complex components
- ▶ Python (for ANSYS and Abaqus) for automating FEA workflows, parametric simulation runs, and post-processing
- ▶ Advanced Additive Manufacturing like FDM, ECP etc

## **ACHIEVEMENTS & RECOGNITION**

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- ▶ **Certified by ANSYS for Fluid Static, and Supersonic Flow.**
- ▶ 3rd Place in CAD Design Competition, IIT Ropar (50+ participants from top engineering colleges)
- ▶ 40+ self-initiated projects/designs spanning physical builds (UAVs, rockets, drones) and digital/CAD/simulation work, all driven by independent curiosity.
- ▶ IDEathon Certificate: CSJM University (Official University Innovation Challenge; idea selected as a significant societal contribution)
- ▶ IDEathon Proposal: Outreach programme to conduct hands-on workshops in Robotics, RC Mechanisms, OTA Control Systems, RC Planes & Drones for Class 8 students, targeting the most curious, eccentric, unburdened young minds.
- ▶ RC Workshop Mentor: CSJM University Kanpur (selected to mentor students in RC aircraft design & fabrication)
- ▶ Workshop Participant: HBTU Kanpur (technical workshops in engineering design and fabrication)

## **A NOTE ON INTENT**

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*I am applying not to showcase what I already know, but because I am VERY keen to learn. Every project listed here was built out of curiosity with no formal guidance, from scratch, through failure, and iteration. I am willing to work hard, absorb everything, and contribute meaningfully. An internship at your institution would be the single most valuable step in my engineering journey.*

## **DECLARATION**

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*I hereby state that all information provided above is true to the best of my knowledge and belief.*

(As of 11 April 2026)

**Shaurya Bhatnagar**